



Python Programming Foundations

Week 7 - RBA & Project Framing
Day 2: "Constraints, Structure, & Scope Control"

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Smaller And Clearer Is Stronger

A smaller clear project usually beats a larger vague project.

The capstone is not judged by how many features you imagine.

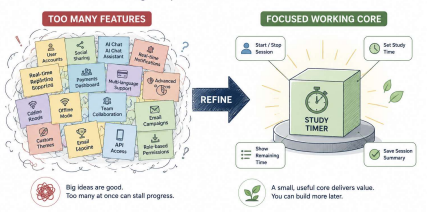
It is judged by whether the project can run, be tested, and be explained.

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IDEA REFINEMENT

Start big to explore. Finish focused to succeed.



TOO MANY FEATURES

- Too many features
- Big ideas are good. Too many at once can stall progress.

FOCUSED WORKING CORE

- A small, useful core delivers value. You can build more later.

REFINE

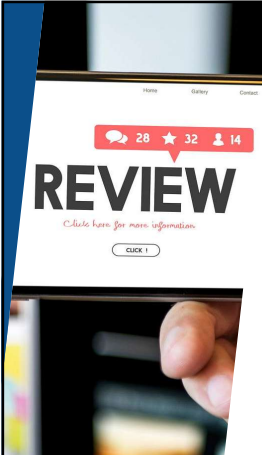
FOCUS CREATES MOMENTUM. Solve one problem well first.

Project Refinement

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Review: Framing Became Raw Material



Your framing work should now help answer:

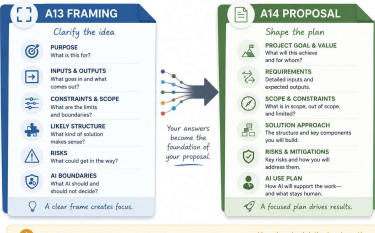
- ▶ What is the project for?
- ▶ What does it need?
- ▶ What does it produce?
- ▶ What limits matter?
- ▶ What should AI not decide?

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FROM FRAMING TO PROPOSAL

Yesterday's clarity becomes tomorrow's plan.



A13 FRAMING

- Clarify the idea
- PURPOSE: What is this for?
- INPUTS & OUTPUTS: What goes in and what comes out?
- CONSTRAINTS & SCOPE: What are the limits, and boundaries?
- LIKELY STRUCTURE: What kind of solution makes sense?
- RISKS: What could get in the way?
- AI BOUNDARIES: What AI should and should not decide?
- A clear frame creates focus.

A14 PROPOSAL

- Shape the plan
- PROJECT GOAL & VALUE: What is the action, and for whom?
- REQUIREMENTS: Detailed inputs and expected outputs.
- SCOPE & CONSTRAINTS: What is in scope, out of scope, and inside?
- SOLUTION APPROACH: The structure and key components you will build.
- RISKS & MITIGATIONS: Key risks and how you will address them.
- AI USE PLAN: How AI will support the work—and what stays human.
- A focused plan drives results.

Your answers become the foundation of your proposal.

Good framing makes proposal writing faster, easier, and stronger. You already did the hard part!

Review: Framing Became Raw Material

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What Counts as Success Today

- Identify the realistic core
- Reduce or park extra features
- Choose a structure that fits the project
- Name risks before coding
- Define AI-use boundaries
- Revise the idea into an approvable proposal

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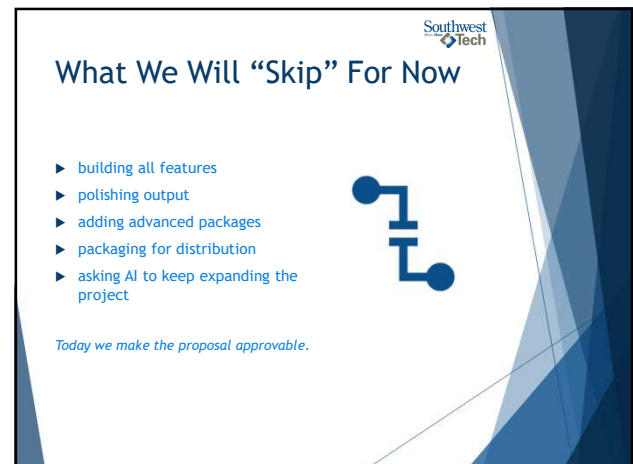
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Constraints Are Part Of The Design

"Good constraints make the project stronger by making reality visible."

- time
- skill level
- data access
- explainability
- testing difficulty
- course fit

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CONSTRAINTS SHAPE CLARITY

Good constraints turn a big idea into a plan you can hit.

Constraints are not limits. They are design guides that help you succeed.

Constraints Are Part Of The Design

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Feature Count Is Not Project Quality

More features can make a project weaker if they make it harder to finish or explain.

Capstone quality comes from a coherent working core.

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MANY SCATTERED FEATURES

Lots of ideas. Hard to build. Hard to explain.

ONE COHERENT WORKING CORE

Focused. Connected. Deliverable.

Scattered focus leads to complexity, higher risk, and unclear priorities.

A coherent core is easier to build, easier to explain, and easier to trust.

Quality comes from a coherent core. Start strong. Build with intention.

Feature Count Is Not Project Quality

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Scope Reduction Is A Professional Move

Scope reduction asks:

- What is the core useful behavior?
- What can be removed?
- What can become a future feature?
- What must remain explainable?

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SCOPE REDUCTION: FOCUS THE CORE

Move what's not essential today into a clear future list.

Focus creates momentum. A smaller, stronger core today makes a bigger impact tomorrow.

Scope Reduction Is A Professional Move

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Structure Should Match The Project

Possible structures:

- ▶ one clear script
- ▶ function-based program
- ▶ file/data utility
- ▶ API-style program
- ▶ simple class-based structure, if justified

The structure should serve the project, not decorate it.

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Structure Should Match The Project Slide

PATH SELECTION
Different paths. Same goal: a working, explainable project.

Pick the path that fits

ONE SCRIPT	FUNCTIONS	FILE/DATA UTILITY	API-STYLE PROGRAM	SIMPLE CLASS STRUCTURE
Everything in one file. Good for simple, straightforward tasks.	Break the work into reusable steps. Increases clarity and reduces repeats.	Read, write, and organize data. Useful for local files and datasets.	Expose capabilities through requests. Good when others or other systems will use it.	Group data and behavior together. Helps as complexity grows.

Choose what helps the project finish and explain well.

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Risks Should Be Named Before Coding

Risks are things that could make the project harder:

- ▶ live API instability
- ▶ unclear data format
- ▶ too many features
- ▶ logic not yet understood
- ▶ AI code the student cannot explain

Naming risk early makes revision easier.

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Risks Should Be Named Before Coding

RISK PLANNING: PREPARE, DON'T PANIC
Good plans name risks early so you can design around them.

PROJECT PLAN
Goal: Build a habit tracker that is useful, clear, and easy to explain.

- 1. CORE FEATURES**
 - Sign up and log in
 - Plan
- 2. DATA & STORAGE**
 - Store habits and progress
 - Keep data consistent
 - Plan
- 3. INSIGHTS**
 - Observe and history
 - Simple summaries
 - Plan
- 4. POLISH & LAUNCH**
 - Basic iterations
 - Prepare for sharing
 - Future

API INSTABILITY
External service may change or fail. Plan the fallback and handle errors gracefully.

UNCLEAR DATA
Data definitions or formats may cause confusion. Plan, write assumptions and validate early.

TOO MANY FEATURES
Scope creep can delay the core. Plan: Focus on the core and post ideas for later.

UNEXPLAINED CODE
All suggestions may be hard to understand or maintain. Plan: Review, simplify, and document clearly.

Risks don't disappear. Planning for them keeps the project solid and the explanation clear.
Name it. Plan for it. Move forward with confidence.

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Demo #1

"Scope Reduction Example"

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What to Watch For

Watch the project shrink and improve.

- ▶ The goal is not to make the idea less interesting.
- ▶ The goal is to make the core project finishable.

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PROJECT SCOPE: FOCUS CREATES IMPACT
From everything on the wishlist to a version that works, wows, and can grow.

BEFORE
BIG PROJECT, MANY FEATURES
Lots to build. Harder to finish and explain.

AFTER
FOCUSED PROJECT, STRONG CORE
Clear value. Easier to finish and explain.

ONE STRONG CORE
Core Purpose
Help users build habits and see real progress.

LAYER LIST
Good ideas to focus on.

You're not lowering the goal.
You're choosing the best path to a finish line that matters.

Demo 1: Scope Reduction Example

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Demo #2

"Structure Options Comparison"

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What to Watch For

Different structures fit different projects.

The best structure is the one that makes the project easier to:

- ▶ Finish
- ▶ Test
- ▶ ... And explain.

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STRUCTURE CHOICES: MANY GOOD PATHS
Different structures fit different projects. Choose what helps you build, explain, and maintain your solution.

SCRIPT	FUNCTIONS	DATA UTILITY	API PATH	CLASS-BASED PATH
Everything is one file. Straightforward and easy to run.	Break work into reusable steps. Improve clarity and reduce repeats.	Focus on reading, writing, and managing data and files.	Expose capabilities through requests. Connect with other systems.	Group data and behavior together. Scale with structure.
Good for: • Simple tasks • Prototypes • One-time tools	Good for: • Repeated logic • Clear organization • Easier testing	Good for: • File and data work • Conversions • Local processing	Good for: • Integrations • Remote access • Service-style apps	Good for: • Complex domains • Many related parts • Long-term growth
Example: data_cleaner.py	Example: calculator.py	Example: report_builder.py	Example: user_service.py	Example: task_manager.py

There is no single "best" structure.
Pick the path that fits your project, your skills, and your goals.

Demo 2: Structure Options Comparison

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Demo #3

"AI Boundary Example"

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What to Watch For

AI boundaries should be specific.

"I used AI" is **too vague**.

A useful boundary says

- ▶ what AI may help with
- ▶ what you must decide
- ▶ and what must be verified.

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AI-USE BOUNDARY WORKSHEET

AI can support your work. You stay in charge of the decisions and the quality.

AI MAY HELP WITH	STUDENT MUST DECIDE	STUDENT MUST VERIFY	STUDENT MUST EXPLAIN
Things that speed up work, spark ideas, or handle tedious jobs.	The core choices that shape the project and its direction.	Everything that affects correctness, reliability, and security.	Your reasoning, your process, and how the solution works.
Examples: - Brainstorming features - Outlining options - Drafting rough requests - Explaining concepts - Finding documentation - Suggesting test ideas	Examples: - Project goals and scope - Which features are in or out - Data and structure choices - Architecture and approach - Trade-offs and priorities - Final design direction	Examples: - Code works as intended - Outputs are correct - Edge cases are handled - Performance is acceptable - Data is accurate and valid - Security and privacy are sound	Examples: - Why this approach fits - How the parts work together - Key decisions and trade-offs - How you tested it - What could be improved - What you learned
My notes:	My notes:	My notes:	My notes:

Your commitment
I will use AI responsibly, make my own decisions, verify my work, and explain it in my own words.

Name: _____ Date: _____

Demo 3: AI Boundary Example

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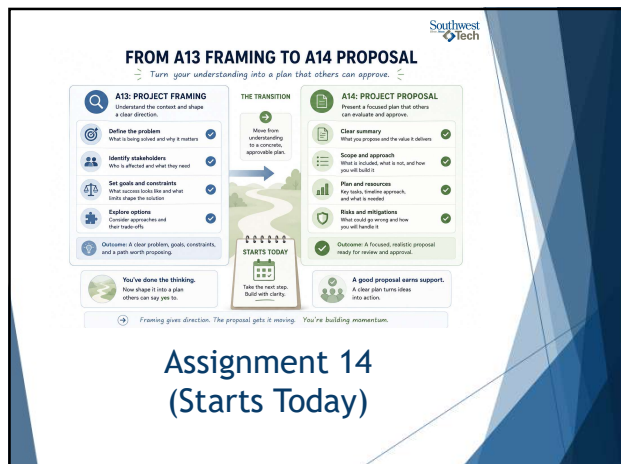
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Assignment #14

Today, Assignment 14 *begins*.
Use your framing work to create a capstone proposal that defines

- ▶ Project name
- ▶ Purpose
- ▶ Features
- ▶ Inputs
- ▶ Outputs
- ▶ Structure
- ▶ Risks
- ▶ AI-use plan
- ▶ Realistic scope.

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Assignment 14 (Starts Today)

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Proposal Evidence

Your proposal should show:

- ▶ clear project purpose
- ▶ feasible feature list
- ▶ expected inputs and outputs
- ▶ likely structure
- ▶ risks or unknowns
- ▶ AI-use boundaries
- ▶ why the scope is realistic

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PROPOSAL CHECKLIST

A clear plan is easier to explain and easier to approve.

1. PURPOSE What problem does this project solve and why does it matter?	☐
2. FEATURES What are the main features? What is in scope and what is out?	☐
3. INPUTS AND OUTPUTS What inputs will the project use? What outputs will it produce?	☐
4. STRUCTURE What approach or structure will you use and why does it fit this project?	☐
5. RISKS What could go wrong? How will you reduce or handle the risks?	☐
6. AI-USE BOUNDARIES Where will AI help? What decisions will you make and verify?	☐
7. REALISTIC SCOPE Can this be built, tested, and explained within the time and resources you have?	☐

Keep it focused. Keep it clear. Keep it doable.
A strong proposal turns ideas into action.

Proposal Evidence

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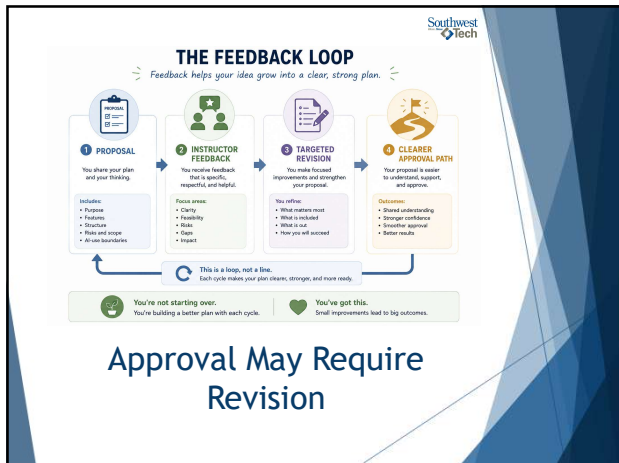
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Approval May Require Revision

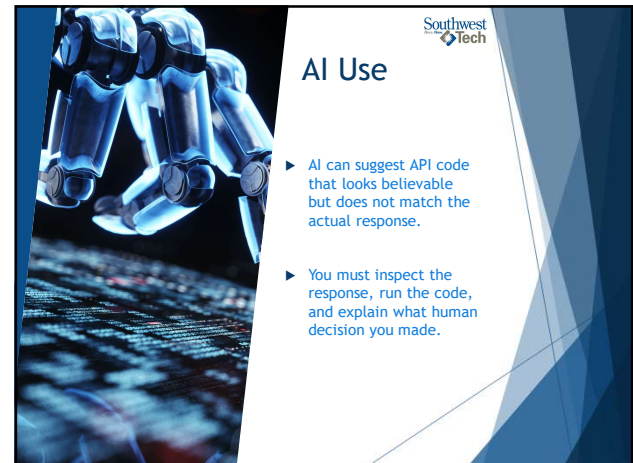
Approval is not just **yes** or **no**.

The instructor may approve, approve with limits, or request targeted revision so the project becomes more realistic.

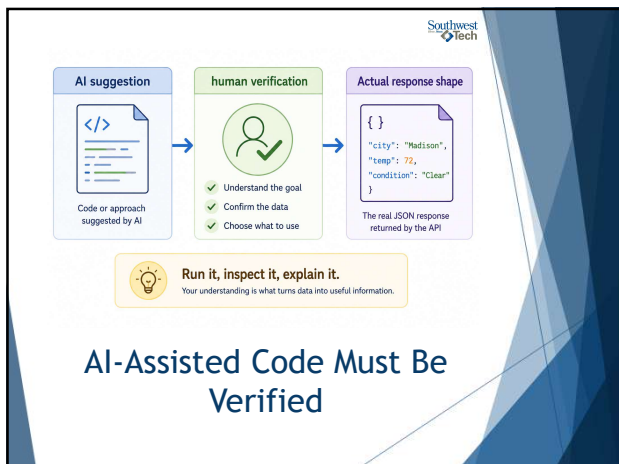
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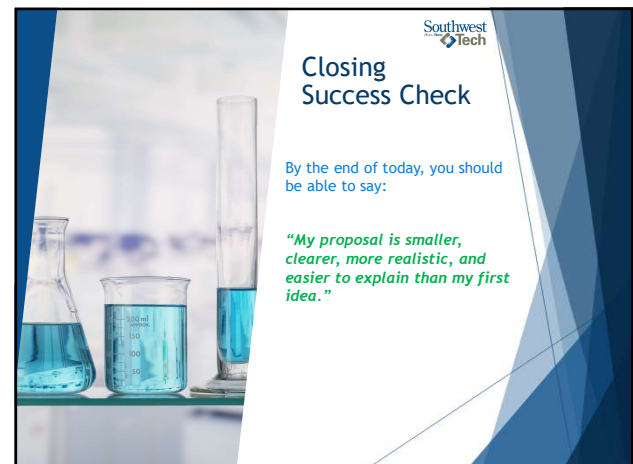
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